RELATIVE PERFECTION OVER A RING TESTED AGAINST ALL PERFECT PAIRINGS, VIA FIBRE BOUNDEDNESS AND A GENERATOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12780-12787 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12789-12829, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 8 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted B side (both complete). Second load-bearing use of wave-17 cards L16169/L16171.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Chapter lemmas characterize-pseudo-coh-improved, relative-pseudo-coherent-is-moot (L16171), bounded-on-fibres (L16169); Derived Categories of Spaces: detecting-bounded-below, compare-base-change, bondal-van-den-Bergh, pullback-generator; Cohomology on Sites, lemma dual-perfect-complex.

Source reference. Original lemma 0GFK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12759–12769. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GES: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12230–12237. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DII: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 10860–10868. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08JJ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12829–12829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08IR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12829–12829. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GFI: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12829–12829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E4R: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12829–12829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09IY: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12829–12829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let A be a ring. Let X be an algebraic space separated, of finite presentation, and flat over A. Let $K \in D_{QCoh}(\mathcal{O}_X)$. If $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is perfect in $D(A)$ for every perfect $E \in D(\mathcal{O}_X)$, then K is* $\mathrm{Spec}(A)$*-perfect.*

Proof. By Lemma 0GES, K is pseudo-coherent relative to A. By Lemma 0DII, K is pseudo-coherent in $D(\mathcal{O}_X)$. By Derived Categories of Spaces, Proposition 0GFI we see that K is in $D^-(\mathcal{O}_X)$. Let $\mathfrak{p}$ be a prime ideal of A and denote $i : Y \to X$ the inclusion of the scheme theoretic fibre over $\mathfrak{p}$, i.e., Y is a scheme over $\kappa(\mathfrak{p})$. By Lemma 0GFK, we will be done if we can show $Li^*(K)$ is bounded below. Let $G \in D_{perf}(\mathcal{O}_X)$ be a perfect complex which generates $D_{QCoh}(\mathcal{O}_X)$, see Derived Categories of Spaces, Theorem 09IY. We have

$$\begin{aligned} R\mathrm{Hom}_{\mathcal{O}_Y}(Li^*(G), Li^*(K)) &= R\Gamma(Y, Li^*(G^\vee \otimes^{\mathbf{L}} K)) \\ &= R\Gamma(X, G^\vee \otimes^{\mathbf{L}} K) \otimes^{\mathbf{L}}_A \kappa(\mathfrak{p}) \end{aligned}$$

The first equality uses that Li^* preserves perfect objects and duals and Cohomology on Sites, Lemma 08JJ; we omit some details. The second equality follows from Derived Categories of Spaces, Lemma 08IR as X is flat over A. It follows from our hypothesis that this is a perfect object of $D(\kappa(\mathfrak{p}))$. The object $Li^*(G) \in D_{perf}(\mathcal{O}_Y)$ generates $D_{QCoh}(\mathcal{O}_Y)$ by Derived Categories of Spaces, Remark 0E4R. Hence Derived Categories of Spaces, Proposition 0GFI now implies that $Li^*(K)$ is bounded below and we win. □